XR School Implemented Simulation Quality Report

Audit date: 2026-08-01

Scope: 37 released simulations

Portfolio average: 79.2/100

Evidence maturity: 37 internal QA; 0 device verified; 0 classroom verified

Audit position: Released means publicly launchable. Released does not mean school-validated, Quest-verified, classroom-verified, or proven to improve learning outcomes.

Executive summary

The released portfolio contains 37 canonical simulations. Its evidence-backed product-indicator average is 79.2/100: 1 pilot candidates, 36 promising internal-QA classes, 0 needing focused improvement, and 0 requiring rebuild before pilot.

Repository evidence records 248 narration cues, 248 packaged narration clips, 0 missing narration files, and 106 declared assets. These are implementation indicators, not learner-outcome measurements.

Quest and classroom evidence are absent: no signed physical-device acceptance runs or controlled classroom studies are represented in this audit. Every class therefore remains at internal QA evidence maturity.

Ranked portfolio

Rank | Simulation | Canonical slug | Score | Band | Evidence maturity

1 | Soluble and Insoluble Substances Lab | c5-ch07-a03-soluble-and-insoluble-substances | 85 | Pilot candidate | internalQA

2 | Living Mycelium Lab: Fungi and Its Development | c8-ch02-a03-fungi-and-its-development | 84 | Promising internal QA | internalQA

3 | Ploughing: Preparation of Soil | c8-ch01-a01-ploughing-preparation-of-soil | 84 | Promising internal QA | internalQA

4 | Plant Pollination & Growth Cycle | pollination | 83 | Promising internal QA | internalQA

5 | Sorting Materials According to Their Shape | c6-ch04-a01-sorting-materials-according-to-their-shape | 82 | Promising internal QA | internalQA

6 | Sources of Vitamins and Their Deficiencies | c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies | 82 | Promising internal QA | internalQA

7 | Test the Presence of Lipids | c6-ch02-a03-test-the-presence-of-lipids | 82 | Promising internal QA | internalQA

8 | The Sources of Minerals in Food | c6-ch02-a05-the-sources-of-minerals-in-food | 82 | Promising internal QA | internalQA

9 | What Floats, What Sinks? | c5-ch07-a01-a-concept-about-what-floats-what-sinks | 82 | Promising internal QA | internalQA

10 | Solar System: Gravity's Orchestra | c8-10-science-solar-system | 81 | Promising internal QA | internalQA

11 | States of Matter Particle Lab | c9-ch01-a02-states-of-matter | 81 | Promising internal QA | internalQA

12 | Electric Circuits & Resistance (Ohm's Law) | circuit | 80 | Promising internal QA | internalQA

13 | A Step Well Structure | c5-ch06-a02-a-step-well-structure | 79 | Promising internal QA | internalQA

14 | A Visit of an Ancient Fort | c5-ch10-a01-a-visit-of-ancient-fort | 79 | Promising internal QA | internalQA

15 | Camp in the Snow | c5-ch09-a03-camp-in-the-snow | 79 | Promising internal QA | internalQA

16 | Cotton Farming | c6-ch03-a01-cotton-farming | 79 | Promising internal QA | internalQA

17 | Dead Sea: Salt Water and Its Effects | c5-ch07-a02-dead-sea-salt-water-and-its-effects | 79 | Promising internal QA | internalQA

18 | Diagnosis of Malaria | c5-ch08-a01-diagnosis-of-malaria | 79 | Promising internal QA | internalQA

19 | Food Spoilage | c5-ch04-a01-food-spoilage | 79 | Promising internal QA | internalQA

20 | Life Cycle of the Mosquito | c5-ch08-a02-life-cycle-of-the-mosquito | 79 | Promising internal QA | internalQA

21 | Milk Spoilage | c5-ch04-a02-milk-spoilage | 79 | Promising internal QA | internalQA

22 | Pitcher Plant - The Insect Hunter | c5-ch05-a01-pitcher-plant-the-insect-hunter | 79 | Promising internal QA | internalQA

23 | River Crossing Adventure | c5-ch09-a01-river-crossing-adventure | 79 | Promising internal QA | internalQA

24 | Rock Climbing | c5-ch09-a02-rock-climbing | 79 | Promising internal QA | internalQA

25 | Seed Dispersal | c5-ch05-a02-seed-dispersal | 79 | Promising internal QA | internalQA

26 | Snow Mountain Climbing | c5-ch09-a04-snow-mountain-climbing | 79 | Promising internal QA | internalQA

27 | The Making of Aam Papad | c5-ch04-a03-the-making-of-aam-papad | 79 | Promising internal QA | internalQA

28 | The Process of Cotton Ginning | c6-ch03-a02-the-process-of-cotton-ginning | 79 | Promising internal QA | internalQA

29 | The Storage of Rainwater | c5-ch06-a01-the-storage-of-rainwater | 79 | Promising internal QA | internalQA

30 | Preposition Adventure | c2-english-ch01-prepositions | 76 | Promising internal QA | internalQA

31 | Sources of Food Sorting Lab | c6-ch01-a01-sources-of-food | 76 | Promising internal QA | internalQA

32 | Acids, Bases & Neutralisation | c10-ch02-a01-introduction-to-acids-and-bases-and-litmus-test | 75 | Promising internal QA | internalQA

33 | Introduction to Money | c1-math-ch01-introduction-to-money | 75 | Promising internal QA | internalQA

34 | Introduction to the Digestive System | c5-ch03-a02-introduction-of-digestive-system | 75 | Promising internal QA | internalQA

35 | The Effects of Force on an Object's Motion and Shape | c8-ch10-a02-the-effects-of-force-on-object-s-motion-and-shape | 75 | Promising internal QA | internalQA

36 | Colour Adventure | c1-art-a01-learning-of-colours | 74 | Promising internal QA | internalQA

37 | The Breathing Process in Human | c7-ch10-a02-the-breathing-process-in-human | 73 | Promising internal QA | internalQA

Portfolio priorities

Priorities are derived from the three lowest average rubric attainment ratios across the complete 37-card dataset.

1. Deployment readiness: portfolio mean 3.0/5. Address the card-level evidence gaps and next actions before raising evidence maturity.
2. Visual and asset quality: portfolio mean 9.4/15. Address the card-level evidence gaps and next actions before raising evidence maturity.
3. Narration and sound: portfolio mean 7.1/10. Address the card-level evidence gaps and next actions before raising evidence maturity.
4. Physical-device acceptance: run the documented Quest comfort, controller, narration, cleanup, and performance checks; current signed run count is 0.
5. Classroom evidence: collect teacher workflow and learner-comprehension evidence without converting internal QA scores into outcome claims; current study count is 0.

Quality cards

Soluble and Insoluble Substances Lab - 85/100

Canonical slug: c5-ch07-a03-soluble-and-insoluble-substances

Route: /simulations/c5-ch07-a03-soluble-and-insoluble-substances

Publication status: released

Evidence maturity: internalQA

Band: Pilot candidate

Audience: Class 5 - Environmental Science, Science

Soluble and Insoluble Substances Lab is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	19	20
Content / scientific integrity	15	15
Learner interactivity	15	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Living Mycelium Lab: Fungi and Its Development - 84/100

Canonical slug: c8-ch02-a03-fungi-and-its-development

Route: /simulations/c8-ch02-a03-fungi-and-its-development

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Biology, Science

Living Mycelium Lab: Fungi and Its Development is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

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Dimension | Score | Maximum

Educational effectiveness | 18 | 20
Content / scientific integrity | 14 | 15
Learner interactivity | 14 | 15
Visual and asset quality | 11 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
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Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 7 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Ploughing: Preparation of Soil - 84/100

Canonical slug: c8-ch01-a01-ploughing-preparation-of-soil

Route: /simulations/c8-ch01-a01-ploughing-preparation-of-soil

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Environmental Science

Ploughing: Preparation of Soil is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	18	20
Content / scientific integrity	14	15
Learner interactivity	14	15
Visual and asset quality	11	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Plant Pollination & Growth Cycle - 83/100

Canonical slug: pollination

Route: /simulations/pollination

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Classes 6-10 - Biology, Environmental Science

A treatment-and-control investigation that links flower structure, pollination, fertilisation, seed formation, and germination through evidence-gated stages.

Dimension	Score	Maximum
Educational effectiveness	18	20
Content / scientific integrity	13	15
Learner interactivity	14	15
Visual and asset quality	11	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Sorting Materials According to Their Shape - 82/100

Canonical slug: c6-ch04-a01-sorting-materials-according-to-their-shape

Route: /simulations/c6-ch04-a01-sorting-materials-according-to-their-shape

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science

Sorting Materials According to Their Shape is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	18	20
Content / scientific integrity	14	15
Learner interactivity	14	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 4 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Sources of Vitamins and Their Deficiencies - 82/100

Canonical slug: c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies

Route: /simulations/c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

Sources of Vitamins and Their Deficiencies is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	18	20
Content / scientific integrity	14	15
Learner interactivity	14	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

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- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Test the Presence of Lipids - 82/100

Canonical slug: c6-ch02-a03-test-the-presence-of-lipids

Route: /simulations/c6-ch02-a03-test-the-presence-of-lipids

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

Test the Presence of Lipids is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	18	20
Content / scientific integrity	14	15
Learner interactivity	14	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Sources of Minerals in Food - 82/100

Canonical slug: c6-ch02-a05-the-sources-of-minerals-in-food

Route: /simulations/c6-ch02-a05-the-sources-of-minerals-in-food

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

The Sources of Minerals in Food is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	18	20
Content / scientific integrity	14	15
Learner interactivity	14	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 4 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

What Floats, What Sinks? - 82/100

Canonical slug: c5-ch07-a01-a-concept-about-what-floats-what-sinks

Route: /simulations/c5-ch07-a01-a-concept-about-what-floats-what-sinks

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science, Science

What Floats, What Sinks? is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

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Dimension | Score | Maximum

Educational effectiveness | 18 | 20
Content / scientific integrity | 14 | 15
Learner interactivity | 14 | 15
Visual and asset quality | 9 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
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Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Solar System: Gravity's Orchestra - 81/100

Canonical slug: c8-10-science-solar-system

Route: /simulations/c8-10-science-solar-system

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Classes 8-10 - Science, Physics, Geography

A concept-rich mission addressing orbit, temperature, scale, comet behavior, and common astronomy misconceptions.

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Dimension | Score | Maximum

Educational effectiveness | 19 | 20
Content / scientific integrity | 15 | 15
Learner interactivity | 15 | 15
Visual and asset quality | 10 | 15
Narration and sound | 4 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 7 | 10
Deployment readiness | 3 | 5
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Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

States of Matter Particle Lab - 81/100

Canonical slug: c9-ch01-a02-states-of-matter

Route: /simulations/c9-ch01-a02-states-of-matter

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 9 - Chemistry, Physics

A particle-model laboratory that makes spacing, motion, attraction, heating, and phase changes observable without presenting the model as molecular footage.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	14	15
Learner interactivity	13	15
Visual and asset quality	11	15
Narration and sound	7	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 4 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Electric Circuits & Resistance (Ohm's Law) - 80/100

Canonical slug: circuit

Route: /simulations/circuit

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Classes 6-10 - Physics

A tested circuit investigation connecting component manipulation, current flow, resistance, and prediction through Ohm's law.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	14	15
Learner interactivity	13	15
Visual and asset quality	10	15
Narration and sound	7	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
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Gaps and risks

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- Asset richness and visual clarity still need a representative low-end device review.
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Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

A Step Well Structure - 79/100

Canonical slug: c5-ch06-a02-a-step-well-structure

Route: /simulations/c5-ch06-a02-a-step-well-structure

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

A Step Well Structure is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

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Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 13 | 15
Visual and asset quality | 9 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
```

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

A Visit of an Ancient Fort - 79/100

Canonical slug: c5-ch10-a01-a-visit-of-ancient-fort

Route: /simulations/c5-ch10-a01-a-visit-of-ancient-fort

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

A Visit of an Ancient Fort is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Camp in the Snow - 79/100

Canonical slug: c5-ch09-a03-camp-in-the-snow

Route: /simulations/c5-ch09-a03-camp-in-the-snow

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Camp in the Snow is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

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Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 13 | 15
Visual and asset quality | 9 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
```

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Cotton Farming - 79/100

Canonical slug: c6-ch03-a01-cotton-farming

Route: /simulations/c6-ch03-a01-cotton-farming

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Environmental Science

Cotton Farming is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Dead Sea: Salt Water and Its Effects - 79/100

Canonical slug: c5-ch07-a02-dead-sea-salt-water-and-its-effects

Route: /simulations/c5-ch07-a02-dead-sea-salt-water-and-its-effects

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Dead Sea: Salt Water and Its Effects is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 13 | 15
Visual and asset quality | 9 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
```

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Diagnosis of Malaria - 79/100

Canonical slug: c5-ch08-a01-diagnosis-of-malaria

Route: /simulations/c5-ch08-a01-diagnosis-of-malaria

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Diagnosis of Malaria is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Food Spoilage - 79/100

Canonical slug: c5-ch04-a01-food-spoilage

Route: /simulations/c5-ch04-a01-food-spoilage

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Food Spoilage is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Life Cycle of the Mosquito - 79/100

Canonical slug: c5-ch08-a02-life-cycle-of-the-mosquito

Route: /simulations/c5-ch08-a02-life-cycle-of-the-mosquito

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Life Cycle of the Mosquito is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Milk Spoilage - 79/100

Canonical slug: c5-ch04-a02-milk-spoilage

Route: /simulations/c5-ch04-a02-milk-spoilage

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Milk Spoilage is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 13 | 15
Visual and asset quality | 9 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
```

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Pitcher Plant - The Insect Hunter - 79/100

Canonical slug: c5-ch05-a01-pitcher-plant-the-insect-hunter

Route: /simulations/c5-ch05-a01-pitcher-plant-the-insect-hunter

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Pitcher Plant - The Insect Hunter is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 13 | 15
Visual and asset quality | 9 | 15
Narration and sound | 8 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
```

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

River Crossing Adventure - 79/100

Canonical slug: c5-ch09-a01-river-crossing-adventure

Route: /simulations/c5-ch09-a01-river-crossing-adventure

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

River Crossing Adventure is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Rock Climbing - 79/100

Canonical slug: c5-ch09-a02-rock-climbing

Route: /simulations/c5-ch09-a02-rock-climbing

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Rock Climbing is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Seed Dispersal - 79/100

Canonical slug: c5-ch05-a02-seed-dispersal

Route: /simulations/c5-ch05-a02-seed-dispersal

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Seed Dispersal is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Snow Mountain Climbing - 79/100

Canonical slug: c5-ch09-a04-snow-mountain-climbing

Route: /simulations/c5-ch09-a04-snow-mountain-climbing

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Snow Mountain Climbing is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Making of Aam Papad - 79/100

Canonical slug: c5-ch04-a03-the-making-of-aam-papad

Route: /simulations/c5-ch04-a03-the-making-of-aam-papad

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

The Making of Aam Papad is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Process of Cotton Ginning - 79/100

Canonical slug: c6-ch03-a02-the-process-of-cotton-ginning

Route: /simulations/c6-ch03-a02-the-process-of-cotton-ginning

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science

The Process of Cotton Ginning is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Storage of Rainwater - 79/100

Canonical slug: c5-ch06-a01-the-storage-of-rainwater

Route: /simulations/c5-ch06-a01-the-storage-of-rainwater

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

The Storage of Rainwater is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	13	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Preposition Adventure - 76/100

Canonical slug: c2-english-ch01-prepositions

Route: /simulations/c2-english-ch01-prepositions

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 2 - English

A spatial language lesson in which object placement makes position words concrete and supports sentence practice.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 11 | 15
Learner interactivity | 14 | 15
Visual and asset quality | 11 | 15
Narration and sound | 4 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 8 | 10
Deployment readiness | 3 | 5
```

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Sources of Food Sorting Lab - 76/100

Canonical slug: c6-ch01-a01-sources-of-food

Route: /simulations/c6-ch01-a01-sources-of-food

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

A classification investigation with targeted feedback that separates plant, animal, and fungal food sources.

Dimension	Score	Maximum
Educational effectiveness	16	20
Content / scientific integrity	12	15
Learner interactivity	13	15
Visual and asset quality	9	15
Narration and sound	8	10
Usability, accessibility, comfort	8	10
Performance and stability	7	10
Deployment readiness	3	5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 4 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Acids, Bases & Neutralisation - 75/100

Canonical slug: c10-ch02-a01-introduction-to-acids-and-bases-and-litmus-test

Route: /simulations/c10-ch02-a01-introduction-to-acids-and-bases-and-litmus-test

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 10 - Chemistry

An experiment-bench sequence connecting litmus, universal indicator, pH classification, and neutralisation.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 14 | 15
Learner interactivity | 14 | 15
Visual and asset quality | 8 | 15
Narration and sound | 4 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 7 | 10
Deployment readiness | 3 | 5
```

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Introduction to Money - 75/100

Canonical slug: c1-math-ch01-introduction-to-money

Route: /simulations/c1-math-ch01-introduction-to-money

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 1 - Mathematics

An age-appropriate progression through Indian coins, notes, value comparison, simple shopping, and memory.

Dimension	Score	Maximum
Educational effectiveness	17	20
Content / scientific integrity	11	15
Learner interactivity	13	15
Visual and asset quality	11	15
Narration and sound	4	10
Usability, accessibility, comfort	8	10
Performance and stability	8	10
Deployment readiness	3	5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Introduction to the Digestive System - 75/100

Canonical slug: c5-ch03-a02-introduction-of-digestive-system

Route: /simulations/c5-ch03-a02-introduction-of-digestive-system

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science, Biology

A ten-stage pathway investigation covering organs, movement, accessory organs, absorption, recap, and healthy-habit transfer.

```
Dimension | Score | Maximum

Educational effectiveness | 18 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 14 | 15
Visual and asset quality | 8 | 15
Narration and sound | 4 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 7 | 10
Deployment readiness | 3 | 5
```

Strengths

- 10 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 10 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Effects of Force on an Object's Motion and Shape - 75/100

Canonical slug: c8-ch10-a02-the-effects-of-force-on-object-s-motion-and-shape

Route: /simulations/c8-ch10-a02-the-effects-of-force-on-object-s-motion-and-shape

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Physics

A deterministic physics investigation of starting, stopping, speeding, redirection, collision, and deformation.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 14 | 15
Learner interactivity | 14 | 15
Visual and asset quality | 8 | 15
Narration and sound | 4 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 7 | 10
Deployment readiness | 3 | 5
```

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Colour Adventure - 74/100

Canonical slug: c1-art-a01-learning-of-colours

Route: /simulations/c1-art-a01-learning-of-colours

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 1 - Art

An early-years recognition and memory journey with large targets, visible feedback, and a canonical release record.

Dimension	Score	Maximum
Educational effectiveness	16	20
Content / scientific integrity	11	15
Learner interactivity	13	15
Visual and asset quality	12	15
Narration and sound	4	10
Usability, accessibility, comfort	8	10
Performance and stability	7	10
Deployment readiness	3	5

Strengths

- 14 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 14 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Breathing Process in Human - 73/100

Canonical slug: c7-ch10-a02-the-breathing-process-in-human

Route: /simulations/c7-ch10-a02-the-breathing-process-in-human

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 7 - Biology

A six-stage physiology lesson connecting airway structure, diaphragm motion, chest volume, and gas exchange.

```
Dimension | Score | Maximum

Educational effectiveness | 17 | 20
Content / scientific integrity | 13 | 15
Learner interactivity | 13 | 15
Visual and asset quality | 8 | 15
Narration and sound | 4 | 10
Usability, accessibility, comfort | 8 | 10
Performance and stability | 7 | 10
Deployment readiness | 3 | 5
```

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Contribution appendix

This appendix maps every PR #8 contribution to its canonical released class. Scores compare the immutable PR head with the integrated internal-QA implementation.

PR slug | Canonical slug | Before | After | Delta | Main remediation | Remaining risk

walls-tell-stories-ancient-fort-visit | c5-ch10-a01-a-visit-of-ancient-fort | 65 | 79 | +14 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-snow-mountain-climbing | c5-ch09-a04-snow-mountain-climbing | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-camp-in-snow | c5-ch09-a03-camp-in-the-snow | 51 | 79 | +28 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-rock-climbing | c5-ch09-a02-rock-climbing | 61 | 79 | +18 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-river-crossing-adventure | c5-ch09-a01-river-crossing-adventure | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

treat-for-mosquitoes-mosquito-life-cycle | c5-ch08-a02-life-cycle-of-the-mosquito | 57 | 79 | +22 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

treat-for-mosquitoes-malaria-diagnosis | c5-ch08-a01-diagnosis-of-malaria | 57 | 79 | +22 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

experiments-with-water-float-or-sink | c5-ch07-a01-a-concept-about-what-floats-what-sinks | 64 | 82 | +18 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

experiments-with-water-dead-sea-salt-water | c5-ch07-a02-dead-sea-salt-water-and-its-effects | 56 | 79 | +23 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

experiments-with-water-soluble-insoluble | c5-ch07-a03-soluble-and-insoluble-substances | 63 | 85 | +22 | Integrated the useful PR experiment as an enhancement of the existing Solubility class, avoiding a duplicate 36th simulation. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

every-drop-counts-rainwater-storage | c5-ch06-a01-the-storage-of-rainwater | 54 | 79 | +25 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

every-drop-counts-stepwell-structure | c5-ch06-a02-a-step-well-structure | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

seeds-and-seeds-seed-dispersal | c5-ch05-a02-seed-dispersal | 54 | 79 | +25 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

seeds-and-seeds-pitcher-plant | c5-ch05-a01-pitcher-plant-the-insect-hunter | 54 | 79 | +25 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

mangoes-round-the-year-aam-papad | c5-ch04-a03-the-making-of-aam-papad | 50 | 79 | +29 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

mangoes-round-the-year-milk-spoilage | c5-ch04-a02-milk-spoilage | 50 | 79 | +29 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

mangoes-round-the-year-food-spoilage | c5-ch04-a01-food-spoilage | 50 | 79 | +29 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

sorting-materials-by-shape | c6-ch04-a01-sorting-materials-according-to-their-shape | 51 | 82 | +31 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

fibre-to-fabric-cotton-farming | c6-ch03-a01-cotton-farming | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

fibre-to-fabric-cotton-ginning | c6-ch03-a02-the-process-of-cotton-ginning | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

components-of-food-mineral-sources | c6-ch02-a05-the-sources-of-minerals-in-food | 58 | 82 | +24 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

components-of-food-vitamins-deficiencies | c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies | 58 | 82 | +24 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

components-of-food-lipid-test | c6-ch02-a03-test-the-presence-of-lipids | 62 | 82 | +20 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

Rubric, evidence method, and limitations

Weighted rubric

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Dimension | Weight

Educational effectiveness | 20
Content / scientific integrity | 15
Learner interactivity | 15
Visual and asset quality | 15
Narration and sound | 10
Usability, accessibility, comfort | 10
Performance and stability | 10
Deployment readiness | 5
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Evidence method

Scores are computed from canonical definitions, declared stage/action/evidence/assessment counts, narration and asset manifests, focused behavior tests, production-build inclusion, browser-contract observations, and immutable PR-head evidence references.

Limitations

- Scores are product indicators, not grades for a contributor and not evidence of measured learning gain.
- Released does not mean school-validated. Publication status and evidence maturity are separate fields.
- Quest and classroom evidence are absent. Browser/build evidence cannot replace physical-headset or teacher-led acceptance.
- Scores must be recalculated after material implementation changes or new signed evidence.